

- 6 A particle  $P$  is projected with speed  $u$  at an angle  $\alpha$  above the horizontal from a point  $O$  on a horizontal plane and moves freely under gravity. The horizontal and vertical displacements of  $P$  from  $O$  at a subsequent time  $t$  are denoted by  $x$  and  $y$  respectively.

(a) Derive the equation of the trajectory of  $P$  in the form

$$y = x \tan \alpha - \frac{gx^2}{2u^2} \sec^2 \alpha. \quad [3]$$

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During its flight,  $P$  must clear an obstacle of height  $h$  m that is at a horizontal distance of 32 m from the point of projection. When  $u = 40\sqrt{2} \text{ m s}^{-1}$ ,  $P$  just clears the obstacle. When  $u = 40 \text{ m s}^{-1}$ ,  $P$  only achieves 80% of the height required to clear the obstacle.

(b) Find the two possible values of  $h$ . [6]

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